Orange Research

Challenge EURO/ROADEF 2026

Keep the flow!

ROADEF
SOCIÉTÉ FRANÇAISE DE RECHERCHE OPÉRATIONNELLE ET D'AIDE À LA DÉCISION

EURO
THE ASSOCIATION OF EUROPEAN OPERATIONAL RESEARCH SOCIETIES

Challenge ROADEF-EURO

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Les éditions du challenge

https ://roadef.org/challenge

Année	Entreprise	Sujet
1999	Bouygues	Inventory Management Problem
2001	CELAR / DGA	Affectation de fréquences
2003	ONERA & CNES	Programmation d'observation satellitaire
2005	Renault	Séquencement de véhicules automobiles
2007	France Télécom	Planification de techniciens télécom
2009	Amadeus	Gestion de perturbations du trafic aérien
2010	EDF	Optimisation de la production d'électricité
2012	Google	Machine Reassignment Problem
2014	SNCF	Gestion du matériel roulant ferroviaire
2016	Air Liquide	Inventory Routing Problem
2018	Saint-Gobain	Cutting Stock Problem
2020	RTE	Planification de la maintenance du réseau
2022	Renault Group	3D Truck Loading Problem
2026	Orange	

Orange - A key player in digital innovation

A workforce of 8,400 people within the Orange Group contributing to Orange Innovation projects, of which 700 researchers, and an international presence allowing to have expertise as close as possible to our clients.

Patent

11 000 patents in the portfolio

- 1st European operator in terms of intellectual property.
- 32nd ranked worldwide among the top 50 5G patent holders.

Research

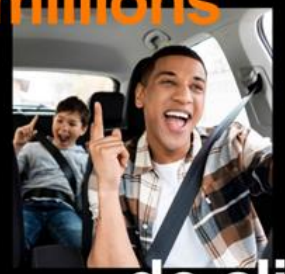
+ 150 active academic and industrial parnerships.
Orange is one of the main private players in digital research in France, promoting value creation.
An approach based on the needs and expectations of users.

Research Objectives :

1. Enlighten and steer the future: identify opportunities and risks
2. Create assets: knowledge & know-how, skills, intellectual property
3. Feeds innovation and value creation: transferring knowledge and assets

Orange - Network and digital services Operator

300 millions



de clients

+2 millions

de clients professionnels et PME

124 600 collaborateurs

Opérateur mobile dans 26 pays

Giving everyone the means to **communicate.**

With high-tech infrastructures deployed from the bottom of the ocean to space, to connect people and companies.

By voice, by image and by data exchange.

+2 500 antennes satellites

45 000 km de fibre terrestre

en Europe, Asie et Etats-Unis

70 000 tours mobiles

dans le monde

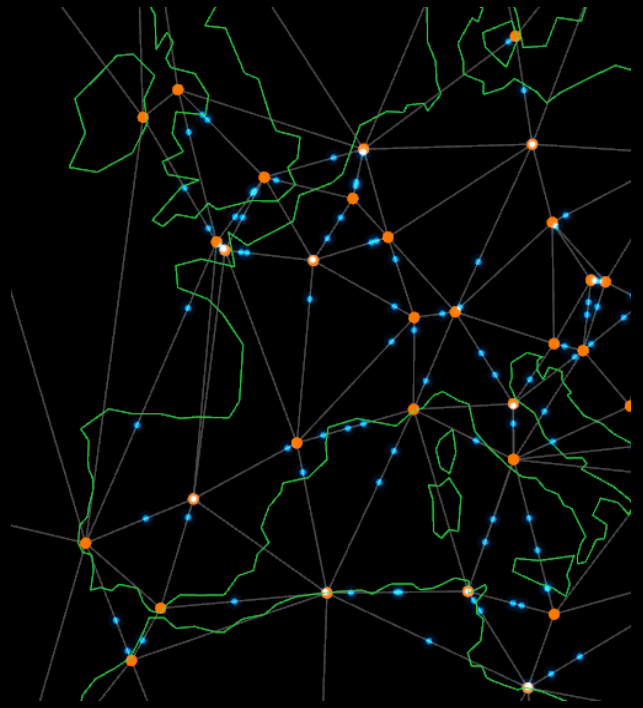
Networks & Traffic Engineering

Telecommunication networks interconnect people, services, and data sources at every moment, across many countries, with guarantees of quality, resilience, and security.

All this traffic flows through shared infrastructures :
- very large Orange core networks = large graphs
- thousands of nodes (POPs)
- tens of thousands of links (10, 40, or 100 Gbps)
- hundreds of thousands of requests (o/d pairs)
- dozens of terabytes/second of transported traffic.

But what does "managing networks" mean ?

1. Ensure a smooth flow of traffic = routing the traffic
2. Design/evolve infrastructures at minimal cost to enable point 1



Routing traffic "optimally" is what we call Traffic Engineering (TE)

"Business" constraints = routing protocols

IP SR ECMP MPLS OSPF BGP

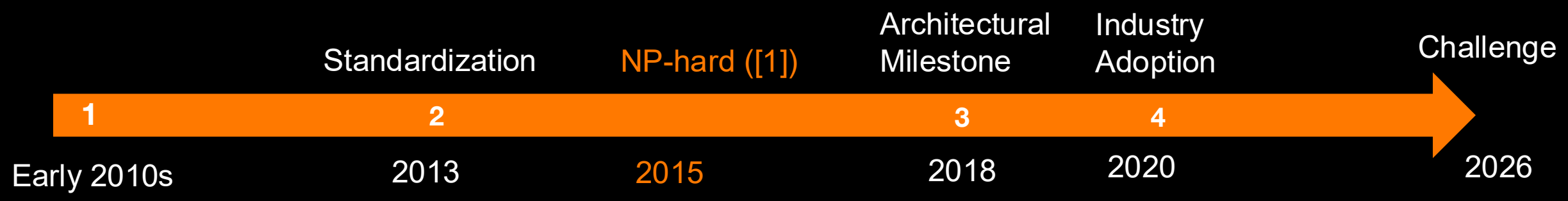
Segment Routing in telecommunication networks

1 - Traditional internet networks relied on shortest-path routing based on fixed link weights.
This approach was limited by the complexity of finding weights avoiding congestion while remaining robust.

2 - The *Internet Engineering Task Force* proposed Segment Routing (SR) to allow packets to be routed over non-shortest paths without requiring extensive modifications to existing networks.

3 - The publication of the formal architecture for Segment Routing. This defined "segment paths" as a sequence of "waypoints" that a packet must visit in order.

4 - Major operators began deploying SR in large-scale national and international networks.
It became a critical field for Traffic Engineering (TE).



[1] Renaud Hartert, Pierre Schaus, Stefano Vissicchio, and Olivier Bonaventure. Solving Segment Routing Problems with Hybrid Constraint Programming Techniques. In Principles and Practice of Constraint Programming. 2015.

Segment Routing VS Shortest Path Routing

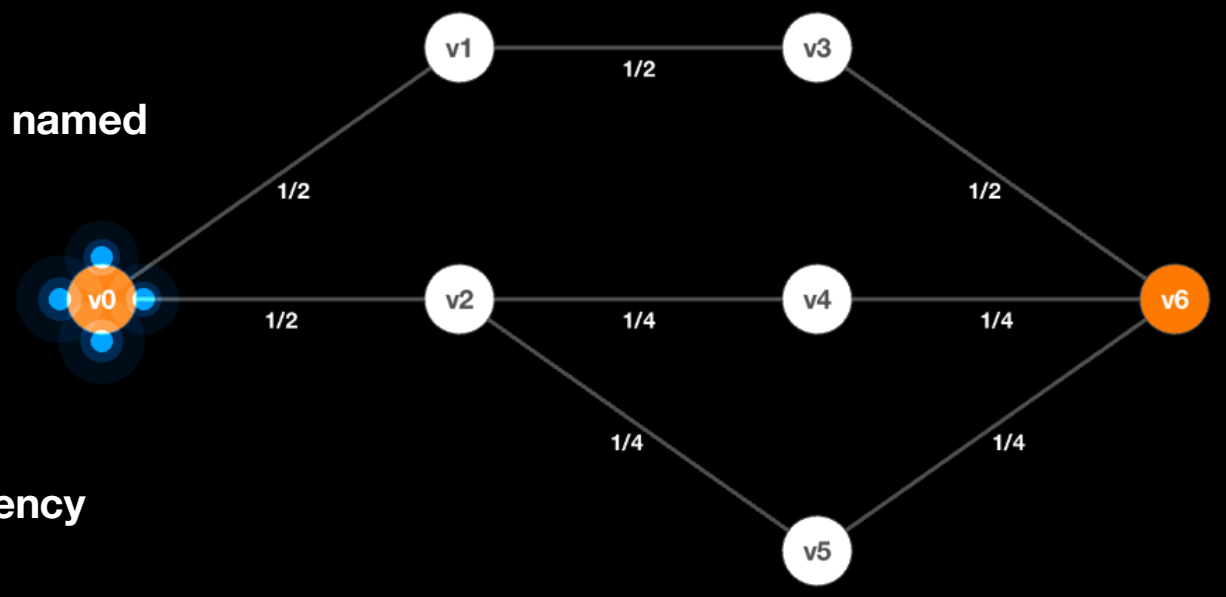
Traditional Routing

Paths are computed via weighted shortest paths (weight named 'metric')

Whenever several shortest paths exist, the ECMP rule applies (Equal-Cost Multi-Paths):

→ traffic is split at junction nodes

Lack of flexibility: congestion, underused links, low resiliency



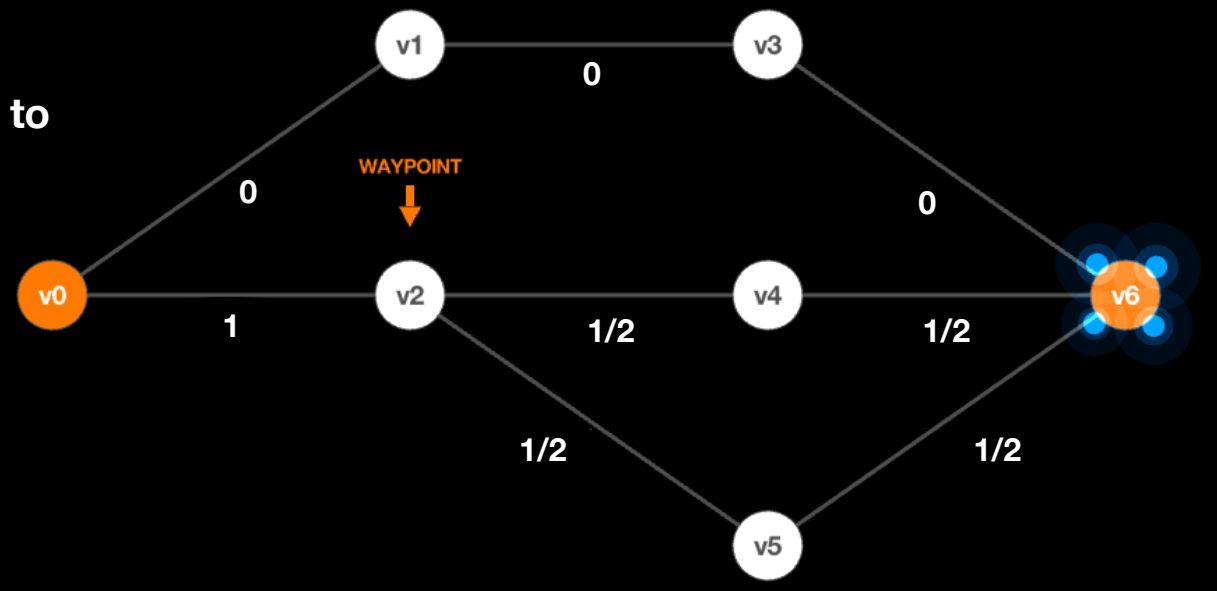
Segment Routing

Intermediate nodes (waypoints) to which all traffic is forced to go through

Shortest path routing between waypoints

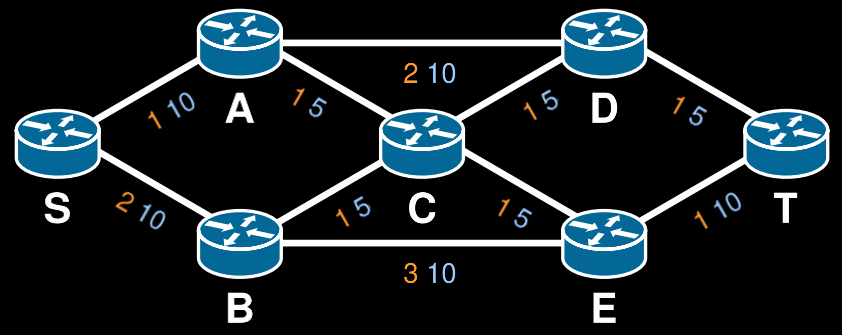
Segment Path ⟨src, w1, w2, …, dst⟩

⟨src, dst⟩ is a possible segment path



Challenge Problem : T-Adaptive Segment Routing Problem (T-ASR)

A network : graph + weights + capacities



A traffic matrix per time step (days, weeks, …)

	S	A	B	T
S				6
A				
B				4
T				

t=0

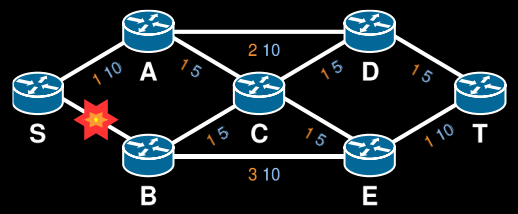
	S	A	B	T
S				2
A				
B				3
T				

t=1

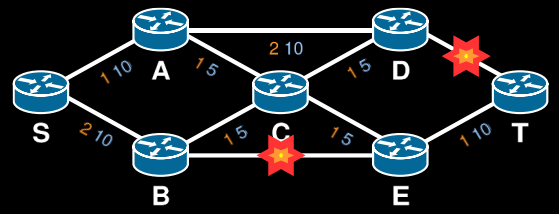
…

An interventions scenario

- At each timestep, some links are turned down



t=3



t=5

A budget at each timestep

- To change the waypoints already set up in the timestep before

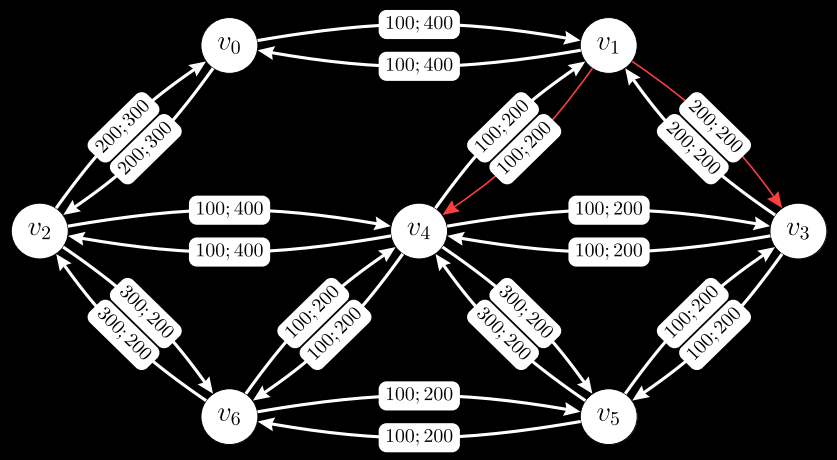
Output

A segment path for each demand at each timestep such that
- The budget is respected
- The load over all arcs and all timesteps is minimized

Example

The 'toy' instance

- Directed graph with 7 nodes and 22 arcs
- Labels of arcs: *weight; capacity*
- Traffic matrix with 2 demands d0 and d1
- Scenario with 2 arcs down at timestep 1
  - Arcs (v1,v4) and (v1,v3)

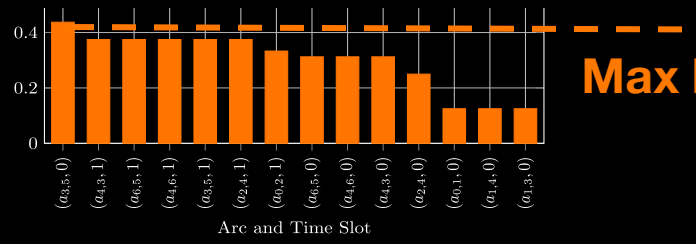


t	d0 (v0,v5)	d1 (v2,v5)
0	50	100
1	100	50

Trivial solution with no waypoints

Demand (v0, v5) uses segment path ⟨v0, v5⟩ for all t

Demand (v2, v5) uses segment path ⟨v2, v5⟩ for all t

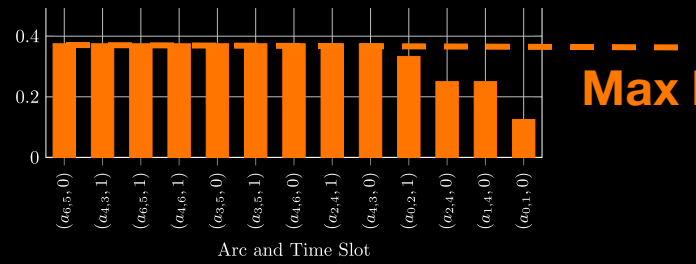


Max load = 0.4375

Optimized solution

Demand (v0, v5) uses segment paths ⟨v0, v4, v5⟩ at t=0
and ⟨v0, v5⟩ at t=1

Demand (v2, v5) uses segment path ⟨v2, v5⟩ for all t



Max load = 0.3750

The T-ASR problem – Formal Definition

Flow conservation constraints : a segment must start where the previous one ended

$$\sum_{j \in V \setminus \{i\}} x_{ij}^{dt} - \sum_{j \in V \setminus \{i\}} x_{ji}^{dt} = \begin{cases} 1 & \text{if } i = s, \\ -1 & \text{if } i = t, \\ 0 & \text{otherwise.} \end{cases} \quad \begin{array}{l} \forall i \in V, \\ \forall d = (s,t) \in D, \forall t \in T \end{array}$$

Maximum number of segments in a Segment Path : maxSeg ≥ 2

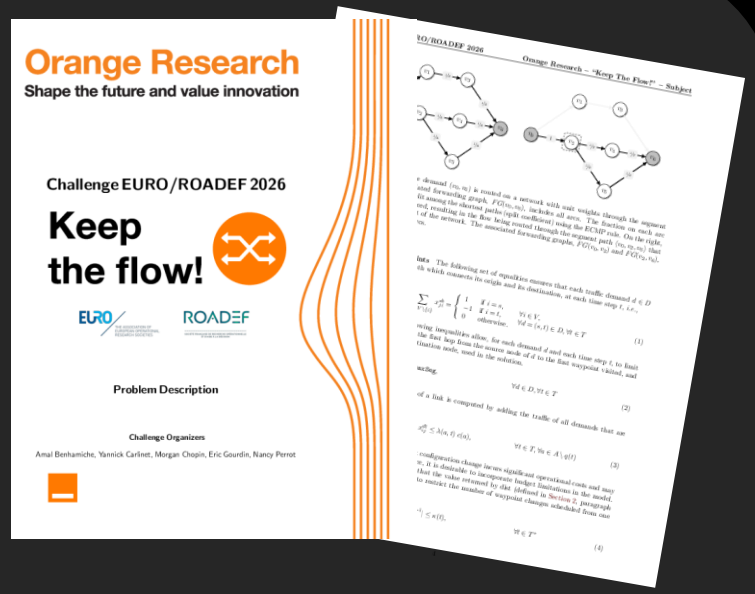
$$\sum_{i,j \in V} x_{ij}^{dt} \leq \texttt{maxSeg}, \qquad \forall d \in D, \forall t \in T$$

Capacity constraints

$$\sum_{d \in D} \sum_{i,j \in V} r(i,j,a,t)\nu(d,t)x_{ij}^{dl} \leq \lambda(a,t)c(a), \quad \forall t \in T, \forall a \in A \setminus q(t)$$

Budget constraints : changing a waypoint from t to t+1 has a cost

$$\sum_{d \in D} \sum_{i,j \in V, i \neq j} |x_{ij}^{dt} - x_{ij}^{d,t-1}| \leq \kappa(t), \qquad \forall t \in T^*$$



Check out the PDF for a proper formal definition of T-ASR

Networktools : A C++20 Network Optimization Library

What ?

An open source C++20 library dedicated to modeling and solving network optimization problems
It is being developed at Orange Research and powers internal decision aiding tools

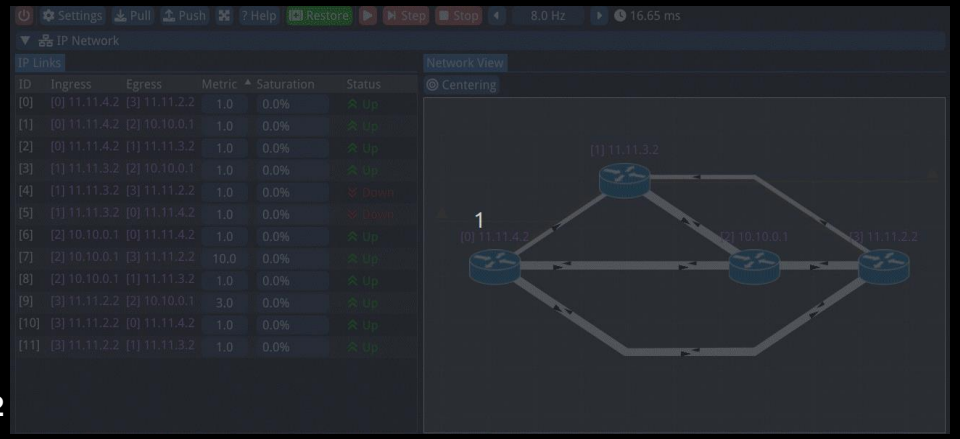
Main features

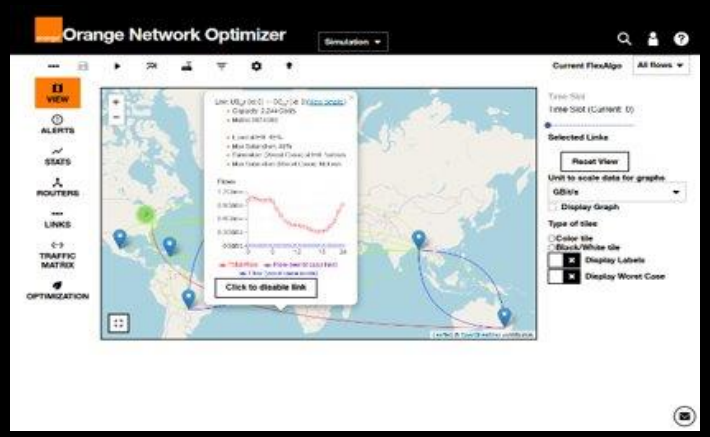
Integrate the C++ LEMON library: wide range of network/graph algorithms

Fast and generic: C++ meta-programming & low-level memory management

Header only and no external dependencies

Unified API for using LP/MIP solvers (Cplex, Gurobi, HiGHS, Cbc, Glpk, etc.)





Networktools : A C++20 Network Optimization Library

networktools

Core
Optimized data structures: *d*-heap, hash maps, dynamic arrays, memory allocator, bitfield, …

Linear Programming
High-level interface to several LP and MIP solvers (CPLEX, GUROBI, HiGHS, CBC, …)

Graph Theory
Cache-optimized graph representation & algorithms

Traffic Engineering (TE)
Routing protocol simulation (OSPF, Segment Routing, ...) and optimization algorithms

```
LocalSearchMetrics ls(network, capacities);

LocalSearchMetrics::Parameters params;
params.setIntegerParam(NUM_ITER, 6);
params.setIntegerParam(SYMMETRIC_WEIGHT, true);
params.setIntegerParam(ROBUST_WEIGHTS, true);
ls.setParameters(params);

ls.setWeightCandidates(6000, 60000, 5);

ls.run(demand_graph, demand_values);
const float f_max_sat = ls.bestScore();
```

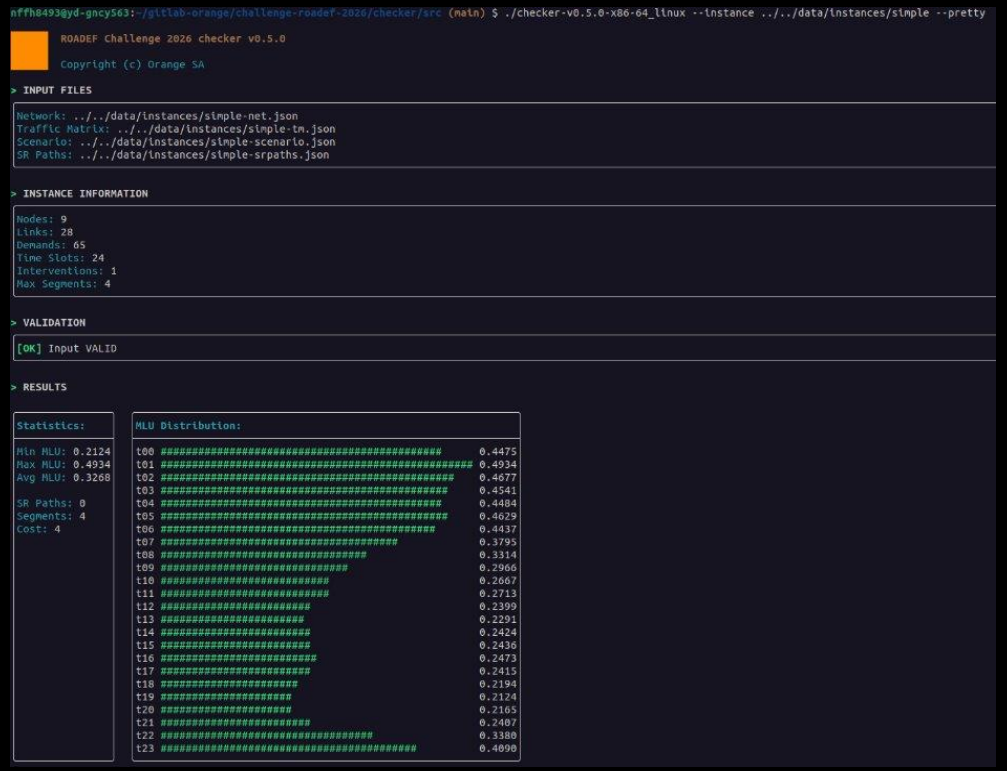
```
SrPath sr_path;
sr_path.addSegment(A);
sr_path.addSegment(E);
sr_path.addSegment(G);

using SR = te::SegmentRouting< Digraph >;
SR sr(network);
sr.routeFlow(sr_path, 10);
```

Checker : Validate And Compute Solution Score

The checker is implemented in C++20 using networktools as a command line tool

```
checker --net <network_file> --tm <traffic_matrix> --scenario <scenario_file>
        --srpaths <solution_file> [--verbose] [--pretty] [--max-decimal-places <n>]
```



Pretty mode (graphic output)

```
INFO| Logging to 'checker.log', mode: 'w', verbosity: 9
INFO| Loading network topology from '../../data/instances/simple-net.json'...
INFO| ✅ Network topology '../../data/instances/simple-net.json' successfully loaded (9 nodes, 28 arcs)
INFO| Loading traffic matrix from '../../data/instances/simple-tm.json'...
INFO| ✅ Demand graph '../../data/instances/simple-tm.json' successfully loaded (65 demands)
INFO| Loading scenario from '../../data/instances/simple-scenario.json'...
 ERR| ❌ Budget timeslot '25' is out of bounds [1,24)
```

Verbose gives more details to the user in case of errors


```
{"valid":true,"total_cost":3,"total_srpaths":1,"total_segments":5,"objectives":[{"t":0.0,"mlu":0.375,"inv_load_c
ost":3.209523809523,"jain_index":0.291919191919},{"t":1.0,"mlu":0.375,"inv_load_cost":3.500000089406,"jain_index
":0.27224268269}],"saturations":[{"t":0,"from":6,"to":5,"sat":0.375},{"t":1,"from":4,"to":3,"sat":0.375},{"t":1,
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.0},{"t":0,"from":0,"to":2,"sat":0.0},{"t":1,"from":6,"to":4,"sat":0.0},{"t":1,"from":6,"to":2,"sat":0.0}]}
```
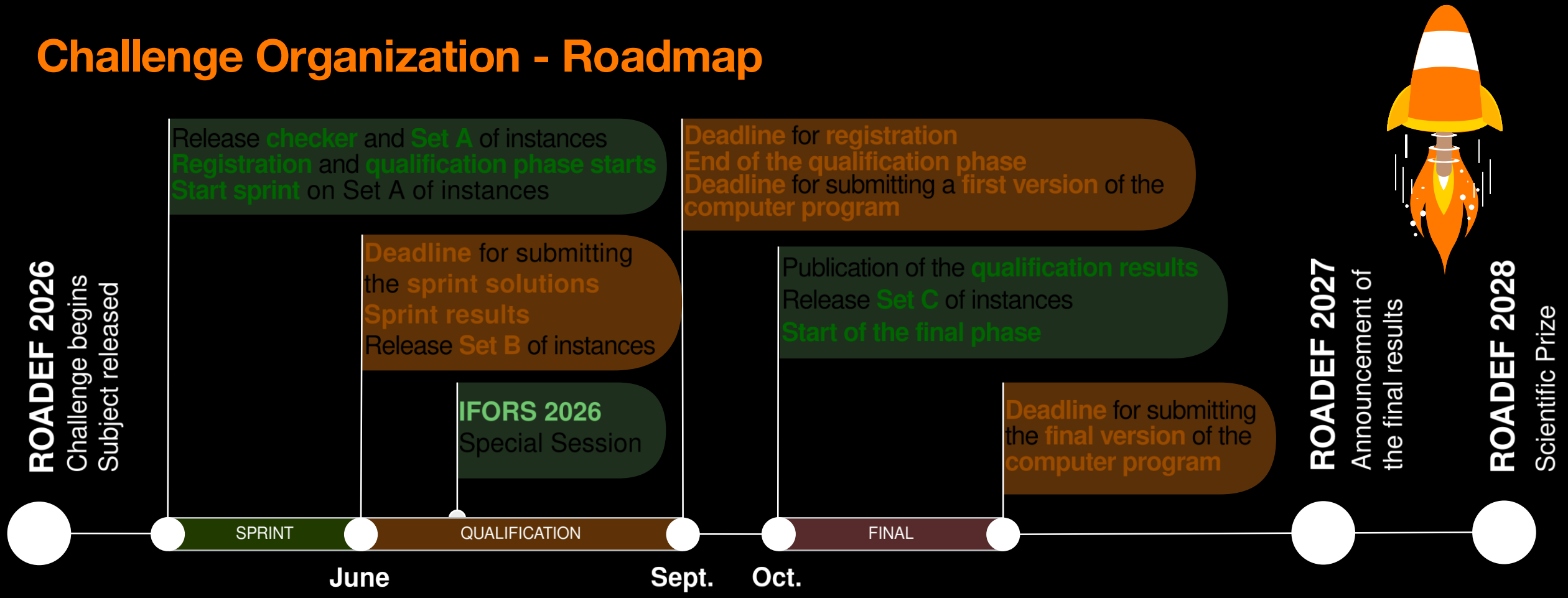


Default execution mode outputs directly the loads and score in JSON

Challenge Organization - Roadmap



Sprint	**March, 6 2026**	➡	**June, 1 2026**
Qualification	**June, 15 2026**	➡	**September, 14 2026**
Final	**October, 5 2026**	➡	**January, 15 2027**

Challenge Organization - Rules

Three datasets

Dataset A : Initial "easy" problem instances, horizon T=2
Dataset B : Intermediate difficulty instances, unbounded horizon
Dataset C + X : Hardest instances (set X disclosed after evaluation)

Each instance is composed of **3 JSON files** : `network.json, tm.json and scenario.json`

A **problem solution** (set of **SR-paths**) must be described in a **JSON file** : `srpaths.json`

Available solvers

Cplex
Gurobi
LocalSolver

Evaluation

Sprint	only solution files
Qualification	a program with short document (2 pages max)
Final	a program with short document (5 pages max)

Check out the PDF for more details about the challenge rules

Challenge Organization - Prizes

Category	Prize	Comment
Sprint best team	2,500 €	
Qualification best team	5,000 €	
Final best team	10,000 €	
Final 2nd best team	5,000 €	
Final 3rd best team	2,500 €	
Junior best team	5,000 €	Composed only of students
Scientific best contribution	5,000 €	Referred by an external committee

→ 35,000 € Total prizes

Conclusion

The challenge starts next week !

All details at https://roadef.org/challenge/2026

Checker and instances:

- https://gitlab.com/Orange-OpenSource/network-optimization-tools/challenge-roadef-2026
- Available on March 6

Networktools:

- https://gitlab.com/Orange-OpenSource/network-optimization-tools/networktools
- Available on March 6

Thank you